



Course Syllabus

IN403 Deep Learning and Artificial Intelligence

COURSE INFORMATION

Course Title:	Deep Learning and Artificial Intelligence
Credit Hours:	6
Prerequisites:	None

COURSE DESCRIPTION

This broader course is a subset of machine learning that will provide an overview of neural networks and advanced architectures including deep neural networks, convolutional networks, and recurrent networks. The course provides an overview of artificial intelligence achievements, including image, voice, and handwriting recognition, and natural language processing using core Python packages based on long short-term memory (LSTM), automatic speech recognition (ASR), and image classification methods (MNIST). Topics covered include unsupervised learning, random forests, and model training.

LEARNING OUTCOMES

By the end of this course, you should be able to:

- IN403-1:** Create deep neural networks for application to the field of information technology.
- IN403-2:** Explore Bayesian Machine Learning.
- IN403-3:** Use a deep learning model to develop solutions to real life problems.
- IN403-4:** Analyze multiple case studies involving deep learning.
- IN403-5:** Develop a deep learning project.
- IN403-6:** Examine risks and controls in deployed AI systems.
- GEL-4.03:** Understand patterns of human behavior based upon real world observation.
- GEL-1.02:** Demonstrate college-level communication through the composition of original materials in Standard English.

COURSE MATERIALS

Access [Instructor Contact Information](#)

For courses with ebooks or digital books, click on the module titled Digital Book in the Table of Contents.

If your course has a physical book that you have not received, please contact your Student Advisor if you have not done so previously.

TEXTBOOK INFORMATION

MediaType: PG Library
Title: Various Library Resources
Edition:
Author: Various
Publisher: Various
Book ISBN:
Ebook ISBN:

MediaType: PG Library
Title: Deep Learning with Python (3rd)
Edition: 3rd (2024)
Author: Watson, M., & Chollet, F.
Publisher: Manning Publications
Book ISBN:
Ebook ISBN: 9781633436589

SOFTWARE REQUIREMENTS

The following software requirements are required in this course beyond the ones listed in the catalog:

No additional technology requirements beyond those in the Purdue Global Catalog.

GRADING CRITERIA/COURSE EVALUATION

Gradebook	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6	Unit 7	Unit 8	Unit 9	Unit 10	Total
Discussion	20	20	20	20	20	20	20	20	20	20	200
Assignments	60	60	60	60	60	60	60	60	60	60	600
Seminar	20	20	20	20	20	20	20	20	20	20	200
Total	100	100	100	100	100	100	100	100	100	100	1000

SPECIFIC PROGRAM/COURSE REQUIREMENTS

Specific Program/Course Requirements will be outlined in your course content.

PURDUE UNIVERSITY GLOBAL GRADING SCALE

Grade	Points	Percent	Grade Point
A	930-1000	93-100%	4.0
A-	900-929	90-92.99%	3.7
B+	870-899	87-89.99%	3.3
B	830-869	83-86.99%	3.0
B-	800-829	80-82.99%	2.7
C+	770-799	77-79.99%	2.3
C	730-769	73-76.99%	2.0
C-	700-729	70-72.99%	1.7
D+	670-699	67-69.99%	1.3
D	600-669	60-66.99%	1.0
F	0-599	Below 60%	0.0

INSTRUCTOR'S GRADING CRITERIA/TIMETABLE

Grades for each unit will be posted by the Monday of the following unit. If that Monday is an observed holiday, grades will be posted by the first business day following the observed holiday.

Late work will be graded within 6 days of the submission date unless that falls on an observed holiday, in which case late work will be graded by the first business day following the observed holiday.

POLICIES

Students who wish to review current policies (academic appeals, attendance/tardiness, etc.) should refer to the [Policy Information](#) in the current Purdue University Global's Catalog.

Purdue University Global's Artificial Intelligence (AI) Policy and Resources can be found on the [PG Center for Teaching and Learning's website](#). Students are encouraged to review the posted policy and refer to course materials for detailed guidance on AI use within each individual course.

As a student, you work hard to earn your degree. Protect the integrity of your work. Do not share your assignments, papers, quizzes, tests, etc. with other students. Do not directly or indirectly share your work by posting it to a third party website. Do not make use of another student's work in any academic activity. Unless specifically instructed to work in groups, do not collaborate with your classmates on assignments or tests. These actions violate the Purdue University Global [Code of Student Conduct](#) policy and, as a result, carry consequences. In some cases, students who are found in violation of these policies are dismissed from the University. You can also review the Writing with Integrity document located in Academic Tools in your course.

If you have questions about these policies, please review these policies in the catalog and contact your instructor.

ACADEMIC ACCOMMODATIONS - If there are any academic accommodations you feel would be helpful to you in your classes, please do not hesitate to reach out to [Student Accessibility Services](#) (SAS) for assistance.

NOTE: The contact information below is to request academic accommodations only and should not be used to contact your instructor. You can contact your instructor through the Brightspace classroom.

Tel: 317-208-1686

Fax: 866-422-4773 (Toll Free)

TTY: TTY users may dial 711 from their TTY phone, at no charge, to be connected with a Telecommunications Relay Services (TRS) operator who will assist with the call.

Accommodations cannot be granted retroactively and you are required to submit supporting documentation needed to process the request.

COURSE SPECIFIC POLICIES

Not Applicable

LATE POLICY

You are expected to submit all assignments and projects by the end of the Unit.

Students may turn in late work up to 2 weeks after the due date. After that, a zero will be assigned for each assignment. Ten percent of the assignment value will be deducted after the assignment is graded for each week that an assignment is late. For example, if your assignment worth 100 points is due on Tuesday and you submit it on Wednesday, 10 points will be deducted after it is graded. If you submit it 2 weeks late, there will be a reduction of 20% of the original assignment value after the assignment is graded.

All late work must be submitted by Sunday of the last unit. No late work will be accepted after this without an [Incomplete Grade](#) approval.

Certain course activities may not be eligible for late submission. These include, but are not limited to, discussions, Seminars, quizzes, labs, exams, and clinical activities.

We know that extenuating circumstances can interfere with timely submission of your work. Be sure to tell your professor and your student advisor right away if you are going to be late with a submission.

Your professor will determine whether an exception to the Purdue Global Late Policy is appropriate. They may also require you to provide documentation of the reason. If your professor determines that an extension is warranted, they will provide you with the expectations for your submission or assignment in writing.

Extenuating circumstances include, but are not limited to, personal/family member hospitalization, family member death, a severe weather event, a natural disaster, and an active military assignment that prevents internet connectivity. Computer-related issues, Internet connectivity, and account blocks are generally not considered extenuating circumstances.

TUTORING

Tutoring and many other resources are available in the Academic Success Center section of the **My Studies** tab on the **Purdue Global Campus home page**. You can also find a link in the Academic Tools within your course.

The Academic Success Center Offers:
Writing help and APA resources
Math help and resources
Business help and resources
Science help and resources
Information Technology help, webinars, and resources

ASSIGNMENTS

Assignments require you to submit coursework to the instructor via the Dropbox. Assignments are graded, whereas learning activities are practice opportunities that are ungraded. Assignments are due Tuesday at 11:59 p.m. ET in the week of their assigned unit. **The course content is outlined within each unit and a calendar is accessible in each course.**

ALL ASSIGNMENTS MUST BE SUBMITTED VIA THE DROPBOX TO ENSURE CREDIT.

Technology Center

The Purdue Global Technology Center offers live tutoring sessions for one-on-one help for working through practice problems, providing additional resources to help you better understand a concept; and helping you load software or troubleshoot errors. The center also

provides online Q&A services when you do not need a live tutor. Questions are answered within 24 hours. Links to the Technology Center are in the online classroom.

Student Conduct

Before completing and submitting assignments, all students should be familiar with and adhere to the [Student Conduct](#) code.

SEMINARS

Participation in the Seminar is an important part of this course. Your weekly participation allows you to interact with your classmates and your instructor in real-time. During the Seminar, you will have the opportunity to review course material in a fun and dynamic forum where you can discuss what you have learned in each unit and ask questions about topics that you do not yet understand.

You must either attend the live Seminar or complete the alternative assignment in order to earn points.

Option 1- Attend Seminar:

Throughout your program, you will be asked to attend various Seminars. Seminars take place at designated times in specific units. Your instructor will give details about the Seminar during the first week of the course. Please review your Course Announcements and emails.

If you are unable to attend the Seminar, you are required to watch the archived recording and complete the alternative assignment by the end of the unit.

Option 2- Alternative Assignment: Reflective Assignment

While listening to this week's archived Seminar, complete a reflective document. You may write the assignment in a bulleted list format OR paragraph format. APA style is not required.

Please include:

- The main topics discussed
- Sub-topics and/or subsequent classroom discussion
- Points you found of interest
- Any additional reflections or suggestions

Keep in mind that your instructor is available should you have any questions on the Seminar content. Submit your alternative assignment to the Seminar Dropbox.

DISCUSSION BOARDS

Discussion topics provide a forum for students to seek clarification and answer important questions about the course material. The discussion also allows students to receive feedback from the instructor and other students in the class. A discussion topic grade will be posted to the Gradebook for each unit using the following grading criteria.

The Discussion Board Requirements

- Post a minimum of 3 posts per discussion topic: one initial response and two replies to your classmates.
- Post on a minimum of 3 different days, for example: Wednesday, Friday, and Monday
- Make the first post by Saturday.

Participating in Discussion Boards

You will communicate with your professor and classmates using the Discussion Board areas. This is where you post your responses to discussion topics and share your experiences in completing exercises and applying the content of the course.

Participation in discussions is required and will help determine your grade. You are expected to contribute to each of the discussion topics in each unit. You should read all responses of other students and add your own.

During discussions, you must demonstrate not only that you understand the subject matter but also that you can apply what you have learned, analyze it, integrate it with knowledge you already have, and evaluate it critically. Your postings should be appropriate and timely contributions that add value to the discussion. Your instructor will provide you more detailed information about requirements and grading related to participation in discussions.

Discussion Guidelines

To ensure your success as a contributor to discussions and as a student in this class, please observe the following guidelines for posting:

- Be clear about which message you are responding to. Refer to specific passages or ideas in the course or text that have sparked your interest.

- Make sure your contribution adds something new to the discussion. A simple "I agree" may be your initial response, but think about how you can take the conversation to the next level.
- Make your posting clear and easy to follow by dividing longer messages into paragraphs.
- Adhere to the Netiquette principles discussed in the following section for ALL Discussion Board communication.
- Address classmates by name or user name, and sign your own messages.
- Feel free to pose new questions to your classmates within your own message.
- Use correct spelling, capitalization, grammar, syntax, and punctuation.
- Communicate your original thoughts. Support your thoughts with properly cited paraphrases and quotations from the course reading and other materials. Plagiarism is totally unacceptable. You are expected to follow APA guidelines for enclosing direct quotations with quotation marks, using in text citations, and listing references.
- If you plan on posting a lengthy response, it is a good idea to type your response in Microsoft® Word® or Notepad™, then copy and paste it into the Discussion Board. This way you will always have a record of your communication if for some reason you lose your connection to the course.
- If you have any questions on these procedures, please contact Student Services.

Sample Topics and Answers

The following are some examples of acceptable and unacceptable responses to discussion topics:

Sample Topic: "The author claims that the internet has changed teaching. How would you respond to this assertion?"

Unacceptable: "I agree."

Unacceptable: "I think the author is off her rocker on this point. I can't stand it when techies try to write about education."

Acceptable: "I agree with Becky W's basic point--that is, I think she's right that the advent of the internet means dramatic changes in how we teach. But I think she's so concerned with whether technology is going to replace teachers that she misses an essential point. Teachers must change themselves to use the internet effectively. The internet will never replace teachers and it will never be a successful tool unless teachers make it one."

Acceptable: "I disagree with Becky W. The internet has not changed teaching any more than the printing press changed teaching, or modern instructional design has changed teaching. New ideas about teaching and new technologies help us deliver good teaching more effectively. They may help us understand more fully what we're delivering. They may help us reach more students. But the essence of good teaching remains the same."

RUBRICS

A rubric is a tool used by your instructor to determine your performance on assessments (assignments, discussions, Seminars, etc.) throughout your course. The rubric lists the criteria the instructor will use to grade your assessment. You should follow the rubric so you understand exactly what is expected of you in each assignment. Following the rubric will help make sure you do not lose points unnecessarily on any graded item. Rubrics are included in the LMS for each Assignment.

NETIQUETTE

Interactions in an online classroom can be in written form as well as in audio or video. Your comfort level with expressing ideas and feelings in writing will add to your success in an online course. The ability to write is necessary, but you also need to understand what is considered appropriate when communicating online.

The word "netiquette" is short for "Internet etiquette." Rules of netiquette have grown organically with the growth of the Internet to help users act responsibly when they access or transmit information online. As a student, you should be aware of the common rules of netiquette for the web and employ a communication style that follows these guidelines.

- Wait to respond to a message that upsets you and be careful of what you say and how you say it.
- Be considerate. Rude or threatening language, inflammatory assertions (often referred to as "flaming"), personal attacks, and other inappropriate communication will not be tolerated.
- Never post a message that is in all capital letters — it comes across to the reader as SHOUTING! Use boldface and italics sparingly, as they can denote sarcasm.
- Keep messages short and to the point.

- Always practice good grammar, punctuation, and composition. This shows that you have taken the time to craft your response and that you respect your classmates' work.
- Keep in mind that Discussion Boards are meant to be constructive exchanges, and it is important to respect those views that are different from yours.
- Be respectful and treat everyone as you would want to be treated yourself.
- Use spell check before sending a written message

Using a webcam in an online meeting room requires thought and consideration for the netiquette environment. Keep your surroundings free of clutter and distraction. Do not allow others in your setting to enter the webcam view. Be sure your attire and background are appropriate for a classroom setting. If in doubt, turn off your camera. When using a microphone be mindful of your mute button - stay on mute until it is an appropriate time to speak - then enable your mute button again when you have finished sharing. Also, be mindful of how you reply to differing opinions or course content that you may not agree with. Feel free to express your views and ask questions in a calm and respectful way.

You should also review and refer to the section on [Responsible Use of University Technology](#) contained in the most recent Purdue Global Catalog.

OTHER POLICIES

Not applicable

Note: This syllabus is subject to change during this term/session or in future terms/sessions. This syllabus was published for course content as of 2602B.